Conformal pullback of FRW type II_∞ classification to spatially curved ($k = \pm 1$) Friedmann–Robertson–Walker backgrounds*

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Abstract

We extend the conformal-pullback type II_∞ classification of [12] (Theorems 3.5/3.6, established for spatially flat $k = 0$ FRW with a comoving doubly-bounded causal diamond and the conformally coupled massless scalar) to spatially curved Friedmann–Robertson–Walker backgrounds. The conformal target is no longer Minkowski but the *Einstein static universe* $(\mathbb{R} \times S^3, g_{\mathrm{ESU}})$ for $k = +1$ or the *Milne chart* $(\mathbb{R} \times H^3, g_{\mathrm{Milne}})$ for $k = -1$. The wave operator identity used in [12] must be replaced by the curvature-corrected form $(\Box_{\tilde{g}} - R[\tilde{g}]/6)(a\varphi) = a^3(\Box_g - R[g]/6)\varphi$ of Wald 1984, App. D.9; we re-verify this identity symbolically in sympy for $k \in \{-1, 0, +1\}$. Two extensions follow. For $k = -1$ (open FRW), the Milne chart is isometric to the interior of the Minkowski future lightcone, so the unitary equivalence and hence the type II_∞ classification of the modular crossed product is *unconditional*. For $k = +1$ (closed FRW), the conformal target is the Einstein static universe; Buchholz–Mund–Summers [2] unconditionally establish the condition of geometric modular action, modular stability and $SO(4,1)$ covariance for transplanted de Sitter wedge nets on RW backgrounds, plus cyclicity on diamonds of sufficient conformal-time extent (their Theorem 3.1, Corollaries 3.2–3.3), but do *not* compute either the type or the Connes spectrum of the ESU spherical-diamond algebra of the conformally coupled scalar. The closed-FRW type II_∞ statement is therefore conditional on the residual hypothesis **(H1$_{+1}$)**: the wedge net of the conformally coupled massless scalar on ESU in its conformally invariant ground state has type III_1 factors with full Connes spectrum on its spherical-diamond subalgebras. We document that BMS [2] themselves flag the conformally-coupled-scalar question on RW (and a fortiori on ESU) as explicitly open in their concluding remarks (p. 16, last paragraph before References).

1 Introduction

The companion note [12] establishes a conformal-pullback type II_∞ classification of the gravitationally dressed algebra of a comoving observer in spatially flat radiation-dominated and matter-dominated FRW (its Theorems 3.5 and 3.6, henceforth "the $k = 0$ theorem" or "Theorem 3.5/3.6" for definiteness): the conformal isometry of $(g_{\mathrm{FRW}}, \eta_{\mu\nu})$ together with conformal invariance of the conformally coupled ($\xi = 1/6$) massless scalar in $d = 4$ lifts to a unitary $U : \mathcal{H}_{\mathrm{FRW}} \to \mathcal{H}_{\mathrm{M}}$ intertwining the FRW conformal vacuum with the Minkowski vacuum and the FRW comoving-diamond

*Standalone companion note to [12], extending its Theorems 3.5/3.6 from spatially flat ($k = 0$) to spatially curved ($k = \pm 1$) FRW. The $k = -1$ extension is unconditional; the $k = +1$ extension is conditional on a residual hypothesis on the conformally coupled massless scalar on the Einstein static universe.

algebra with the Hislop–Longo [7] Minkowski-diamond algebra. By Tomita–Takesaki uniqueness and the Connes–Takesaki crossed-product construction [3, 4], $\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}} \rtimes_{\sigma^{\mathrm{FRW}}} \mathbb{R}$ is a type II_∞ factor whenever $a(\eta)$ is smooth and strictly positive on the closed diamond.

The construction in [12] is restricted to $k = 0$ for two reasons. (a) Conformal flatness of FRW is automatic for $k = 0$ in conformal coordinates; for $k = \pm 1$ the metric is conformally flat as a *Lorentzian* manifold (it is conformal to a chart of either the Einstein static universe or the Milne chart of H^3), but the conformal target is no longer Minkowski. (b) The off-shell wave-operator identity (eq. (5) of [12]) $\Box_{\tilde{g}}(a\varphi) = a^3(\Box_g - R[g]/6)\varphi$ is correct as written only when $R[\tilde{g}] = 0$, i.e. when the conformal target is Ricci-flat; for the Einstein static universe $R[\tilde{g}] = +6$ and for the Milne chart $R[\tilde{g}] = -6$, and an additional curvature term must be carried.

The present note records the extension of [12] to the two remaining cases $k = \pm 1$. Section 2 reviews the spatially curved FRW geometries and the static conformal targets ESU and Milne; Section 3 states and sympy-verifies the curvature-corrected identity (Wald 1984 App. D.9 [14]); Section 4 gives the unconditional $k = -1$ theorem; Section 5 the conditional $k = +1$ theorem with the verbatim Buchholz–Mund–Summers [2] partial resolution; Section 6 the open questions.

Statements (informal). Theorem 2 (open FRW, $k = -1$): *unconditional*, the modular crossed product is a type II_∞ factor. Theorem 9 (closed FRW, $k = +1$): *conditional* on the residual hypothesis $\mathrm{H1}_{+1}$ (type III_1 + full Connes spectrum on the ESU spherical-diamond algebra of the conformally coupled scalar); items (1)–(2) (geometric modular action, modular stability, $SO(4,1)$ covariance, cyclicity on long-enough diamonds) are *unconditional* via BMS [2] Thm. 3.1 and Cors. 3.2–3.3, and only step (3) (crossed-product type) remains open.

Scope and limitations. As in [12]: conformal coupling $\xi = 1/6$, smooth strict positivity of $a(\eta)$ on the closed comoving diamond (bounded away from any zero of a and from any conformal boundary, in particular from Big Bang and Big Crunch); the wedge case is excluded as in [12, Rem. 3.3]; massive, non-conformal ($\xi \neq 1/6$) and interacting fields are out of scope.

2 Preliminaries: spatially curved FRW and conformal targets

2.1 Spatially curved FRW in conformal coordinates

Let $S_k(\chi) := \sin\chi$ for $k = +1$, $S_0(\chi) := \chi$ for $k = 0$, and $S_{-1}(\chi) := \sinh\chi$ for $k = -1$. In conformal time η and hyperspherical comoving coordinates (χ, θ, ϕ), the spatially curved FRW metric reads

$$\mathrm{d}s^2_{\mathrm{FRW}} = a(\eta)^2\Big[-\mathrm{d}\eta^2 + \mathrm{d}\chi^2 + S_k(\chi)^2\big(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2\big)\Big]. \tag{1}$$

The case $k = 0$ recovers [12, eq. (1)]. The conformal-time range is $\eta \in (0, \infty)$ for $k = 0, -1$ and $\eta \in (0, \eta_*)$ for $k = +1$ with $\eta_* = \pi$ (radiation-dominated) or 2π (matter-dominated). For $k = +1$, the Big Bang is at $\eta = 0$ and the Big Crunch at $\eta = \eta_*$. For $k = -1$, the Big Bang is at $\eta = 0$ and the conformal-time range extends to $\eta \to \infty$. Conformal flatness of (1) as a Lorentzian manifold (vanishing Weyl tensor) holds for all k [9, 11, 15].

2.2 Conformal target spacetimes

Writing $g_{\mathrm{FRW}} = a(\eta)^2\,\tilde{g}_k$, the static conformal target $\tilde{g}_k$ is a function of k only:

$$\tilde{g}_0 = -\mathrm{d}\eta^2 + \mathrm{d}\chi^2 + \chi^2\,\mathrm{d}\Omega^2 \;=\; \eta_{\mu\nu} \quad (\text{Minkowski}, R[\tilde{g}] = 0), \tag{2}$$

$$\tilde{g}_{+1} = -\mathrm{d}\eta^2 + \mathrm{d}\chi^2 + \sin^2\chi\,\mathrm{d}\Omega^2 \;=\; g_{\mathrm{ESU}} \quad (\text{Einstein static universe } \mathbb{R}\times S^3,\ R[\tilde{g}] = +6), \tag{3}$$

$$\tilde{g}_{-1} = -\mathrm{d}\eta^2 + \mathrm{d}\chi^2 + \sinh^2\chi\,\mathrm{d}\Omega^2 \;=\; g_{\mathrm{Milne}} \quad (\text{Milne chart } \mathbb{R}\times H^3,\ R[\tilde{g}] = -6). \tag{4}$$

Eq. (2) is the $k = 0$ case used in [12]. Eq. (3) is the conformal-compactification target of any closed FRW [15, §5.4], [11, Ch. 9]. Eq. (4) is the conformal target of any open FRW; it is isometric to the interior of the Minkowski future lightcone $I^+(0) \subset \mathbb{R}^{1,3}$ via $t = e^\eta \cosh\chi$, $r = e^\eta \sinh\chi$ [15, §5.3], so $k = -1$ factors through a further conformal isometry to a subset of $\mathbb{R}^{1,3}$.

Comoving worldline in the target. A comoving observer at $\chi = 0$ in $(g_{\mathrm{FRW}}, k = \pm 1)$ pulls back, under the rescaling $\varphi \mapsto a\varphi$ (Section 3), to the timelike geodesic at $\chi = 0$ in $\tilde{g}_k$: a static observer of ESU for $k = +1$, and a static observer of Milne (which corresponds to the apex of the Minkowski lightcone in the further isometric embedding) for $k = -1$.

3 Curvature-corrected conformal-covariance identity

The off-shell conformal-covariance identity used in [12, eq. (5)], $\Box_{\tilde{g}}(a\varphi) = a^3(\Box_g - R[g]/6)\varphi$, holds when the conformal target $\tilde{g}$ is Ricci-flat ($R[\tilde{g}] = 0$), which is the case for spatially flat FRW. For $k = \pm 1$ the static target $\tilde{g}_k$ is *not* Ricci-flat, and the correct off-shell identity (Wald 1984, Appendix D, eq. D.9 [14]) carries a curvature term:

$$\left(\Box_{\tilde{g}} - \tfrac{1}{6}R[\tilde{g}]\right)(a\,\varphi) \;=\; a^3\left(\Box_g - \tfrac{1}{6}R[g]\right)\varphi. \tag{5}$$

For $k = 0$, $R[\tilde{g}] = 0$ and (5) reduces to the form used in [12]. For $k = +1$, $R[\tilde{g}_{+1}] = R[g_{\mathrm{ESU}}] = +6$; for $k = -1$, $R[\tilde{g}_{-1}] = R[g_{\mathrm{Milne}}] = -6$. (Throughout, we use units in which the spatial unit-curvature radius is 1.)

Sympy verification. The script `piste_spatially_curved.py` (bundled with the source repository, see also [12]) computes both sides of (5) in $(\eta, \chi, \theta, \phi)$ components for arbitrary smooth $a(\eta)$ and arbitrary $\varphi(\eta, \chi, \theta, \phi)$, and reports a difference of zero after `simplify` for all three values $k \in \{-1, 0, +1\}$. The Ricci scalars (re-derived sympy-symbolically from the Christoffel symbols of g_{FRW} and $\tilde{g}_k$ separately) match the analytic expressions $R[g_{\mathrm{FRW}}] = 6\,a^{-2}\left(a''/a + k\right)$ and $R[\tilde{g}_k] = 6k$ exactly.

Consequence. The conformally coupled massless scalar $(\Box_g - R/6)\varphi = 0$ on g_{FRW} pulls back via $\varphi \mapsto \tilde{\varphi} := a\varphi$ to *the conformally coupled massless scalar on the static target* $\tilde{g}_k$, namely $(\Box_{\tilde{g}_k} - R[\tilde{g}_k]/6)\tilde{\varphi} = 0$, *not* to the free wave equation (except for $k = 0$, where $R[\tilde{g}_0] = 0$ and the conformally coupled equation coincides with the free wave equation). In particular, the target theory is the conformally coupled massless scalar on ESU (for $k = +1$) or on Milne (for $k = -1$).

Symplectic form pulls back exactly. The Klein–Gordon symplectic form on a constant-η Cauchy slice $\Sigma_\eta \subset D_{\mathcal{O}}$ (with the curvature term $-(R/6)\varphi^2$ contributing nothing to the antisymmetric current, as in [12, §3.1(b)]) reads, for the spatially curved metric (1),

$$\sigma_{\mathrm{FRW}}(\varphi_1, \varphi_2) \;=\; \int_{\Sigma_\eta} \left(\varphi_1\,\partial_\eta\varphi_2 - \varphi_2\,\partial_\eta\varphi_1\right) a(\eta)^2\, S_k(\chi)^2 \sin\theta\, \mathrm{d}\chi\, \mathrm{d}\theta\, \mathrm{d}\phi, \tag{6}$$

with weight a^2 from $\sqrt{h}\,n^\eta = a^3 \cdot a^{-1} = a^2$, and $S_k(\chi)^2 \sin\theta$ the spatial volume element in $\tilde{g}_k$. Substituting $\varphi_i = \tilde{\varphi}_i/a$ and using $\partial_\eta(\tilde{\varphi}/a) = a^{-1}\partial_\eta\tilde{\varphi} - a^{-2}(\partial_\eta a)\tilde{\varphi}$ as in [12, §3.1(b)], the two $(\partial_\eta a)/a$ cross terms cancel by antisymmetry of the symplectic bracket and one recovers $\sigma_{\mathrm{FRW}}(\varphi_1, \varphi_2) = \sigma_{\tilde{g}_k}(\tilde{\varphi}_1, \tilde{\varphi}_2)$.

4 Theorem (open FRW, $k = -1$): unconditional type II_∞

We now state and prove the unconditional extension of [12, Thm. 3.5] to open FRW, by composition with the classical conformal isometry Milne $\hookrightarrow I^+(0) \subset \mathbb{R}^{1,3}$.

Proposition 1 (Conformal unitary, $k = -1$)**.** *Let $a : [\eta_i, \eta_f] \to \mathbb{R}_+$ be smooth and strictly positive on the closed conformal-time interval, with $0 < \eta_i < \eta_f < \infty$. Let $D_{\mathcal{O}} \subset (\mathrm{FRW}, k = -1)$ be a doubly-bounded causal diamond of a comoving observer with conformal-time interval contained in $[\eta_i, \eta_f]$, and let $D_{\mathcal{O}}^{\mathrm{Milne}}$ denote the same coordinate region viewed as a subset of the Milne chart $\tilde{g}_{-1}$. Then there exists a unitary $U_{-1} : \mathcal{H}_{\mathrm{FRW}} \to \mathcal{H}_{\mathrm{Milne}}$ such that*

(i) $U_{-1}\,\Omega_{\mathrm{FRW}} = \Omega_{\mathrm{Milne}}$, where Ω_{Milne} is the conformally invariant ground state of the conformally coupled massless scalar on $\tilde{g}_{-1}$;

(ii) $U_{-1}\,\varphi_{\mathrm{FRW}}(f)\,U_{-1}^{-1} = \tilde{\varphi}_{\mathrm{Milne}}(a^3 \cdot f)$ for all $f \in C_c^\infty(D_{\mathcal{O}})$, where $\tilde{\varphi}_{\mathrm{Milne}}$ is the rescaled field on Milne;

(iii) $U_{-1}\,\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}}\,U_{-1}^{-1} = \mathcal{A}(D_{\mathcal{O}}^{\mathrm{Milne}})_{\mathrm{Milne}}$ as von Neumann subalgebras of $B(\mathcal{H}_{\mathrm{Milne}})$.

Proof sketch. By (5), the field rescaling $\varphi \mapsto \tilde{\varphi} = a\varphi$ maps the conformally coupled scalar EOM on g_{FRW} to the conformally coupled scalar EOM on $\tilde{g}_{-1}$. By the symplectic computation of (6), the test-function rescaling $f \mapsto a^3 f$ matches the FRW symplectic form to the Milne symplectic form on solutions of $D_{\mathcal{O}}$ and $D_{\mathcal{O}}^{\mathrm{Milne}}$. Smoothness and strict positivity of a on $\overline{D_{\mathcal{O}}}$ make $f \mapsto a^3 f$ a C_c^∞-bijection (its inverse $h \mapsto a^{-3}h$ is also C_c^∞). The induced symplectic transformation lifts to a unitary U_{-1} on the symmetric Fock spaces with $U_{-1}\Omega_{\mathrm{FRW}} = \Omega_{\mathrm{Milne}}$, exactly as in [12, Prop. 3.4]. The double-commutant identity (iii) then follows from (ii) by the same argument. □

Theorem 2 (Open FRW, $k = -1$, unconditional)**.** *Let φ be the conformally coupled massless scalar field on open FRW with smooth strictly positive scale factor $a(\eta)$ on the closed conformal-time interval $[\eta_i, \eta_f]$, $0 < \eta_i < \eta_f < \infty$, in the conformal vacuum Ω_{FRW}. Let $D_{\mathcal{O}}$ be a doubly-bounded causal diamond of a comoving observer with conformal-time interval contained in $[\eta_i, \eta_f]$, and assume the diamond's image $D_{\mathcal{O}}^{\mathrm{Milne}}$ in the Milne chart, lifted under the isometric embedding $\mathrm{Milne} \hookrightarrow I^+(0) \subset \mathbb{R}^{1,3}$, is a doubly-bounded causal diamond $M_{\mathcal{O}} \subset \mathbb{R}^{1,3}$ contained strictly inside the Minkowski future lightcone of the origin (i.e. $\overline{M_{\mathcal{O}}} \cap \partial I^+(0) = \emptyset$). Let σ^{FRW} denote the modular automorphism group of $(\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}}, \Omega_{\mathrm{FRW}})$. Then:*

1. *$\sigma_t^{\mathrm{FRW}} = \mathrm{Ad}\,U_{-1}^{-1} \circ \sigma_t^{\mathrm{HL}} \circ \mathrm{Ad}\,U_{-1}$, where σ^{HL} is the Hislop–Longo modular automorphism of the conformally identified Minkowski diamond $\mathcal{A}(M_{\mathcal{O}})_{\mathrm{M}}$.*

2. *$\Gamma(\sigma^{\mathrm{FRW}}) = \mathbb{R}$.*

3. *$\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}} \rtimes_{\sigma^{\mathrm{FRW}}} \mathbb{R}$ is a type II_∞ factor.*

Proof. By Proposition 1, U_{-1} implements a W^*-isomorphism $\mathrm{Ad}\,U_{-1} : \mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}} \xrightarrow{\sim} \mathcal{A}(D_{\mathcal{O}}^{\mathrm{Milne}})_{\mathrm{Milne}}$ with $U_{-1}\Omega_{\mathrm{FRW}} = \Omega_{\mathrm{Milne}}$. The Milne chart is isometric to $I^+(0) \subset \mathbb{R}^{1,3}$ via $(t, r) = (e^\eta \cosh\chi, e^\eta \sinh\chi)$ [15, §5.3], and the conformally coupled massless scalar is conformally invariant; hence $\mathcal{A}(D_{\mathcal{O}}^{\mathrm{Milne}})_{\mathrm{Milne}} \cong \mathcal{A}(M_{\mathcal{O}})_{\mathrm{M}}$ via a further unitary $V_{-1} : \mathcal{H}_{\mathrm{Milne}} \to \mathcal{H}_{\mathrm{M}}$ sending Ω_{Milne} to Ω_{M} (the embedding sends the Milne ground state to the restriction of the Minkowski vacuum to the lightcone interior, in the conformally coupled sector). The condition $\overline{M_{\mathcal{O}}} \cap \partial I^+(0) = \emptyset$ ensures that the Minkowski-vacuum two-point function and its Hadamard parametrix on $M_{\mathcal{O}}$ are uniformly bounded, so the restriction is well-defined and the unitary V_{-1} exists; this is the Hadamard / Reeh–Schlieder infrastructure of [8, 13]. Composing, $U := V_{-1} \circ U_{-1} : \mathcal{H}_{\mathrm{FRW}} \to \mathcal{H}_{\mathrm{M}}$ implements $\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}} \xrightarrow{\sim} \mathcal{A}(M_{\mathcal{O}})_{\mathrm{M}}$ with $U\Omega_{\mathrm{FRW}} = \Omega_{\mathrm{M}}$. By Hislop–Longo [7], $\mathcal{A}(M_{\mathcal{O}})_{\mathrm{M}}$ in Ω_{M} is a type III_1 factor with Möbius modular flow having full Connes spectrum $\Gamma(\sigma^{\mathrm{HL}}) = \mathbb{R}$. By Tomita–Takesaki uniqueness (Bratteli–Robinson II, §2.5; Takesaki II, Ch. VIII), modular operators intertwine: $\Delta_{\mathrm{FRW}} = U^{-1}\Delta^{\mathrm{HL}}U$, yielding (1). By W*-isomorphism invariance of the Connes spectrum ([12, Prop. 3.4], equivalent to Connes [4,

Théorème 1.2.4]), $\Gamma(\sigma^{\mathrm{FRW}}) = \Gamma(\sigma^{\mathrm{HL}}) = \mathbb{R}$, yielding (2). By Connes–Takesaki, type III_1 crossed by its modular automorphism group with full Connes spectrum $\Gamma = \mathbb{R}$ is type II_∞ [3,4,16, §2], yielding (3). □

Remark 3. The doubly-bounded condition $\overline{M_{\mathcal{O}}} \cap \partial I^+(0) = \emptyset$ is the natural one for any doubly-bounded comoving diamond not touching the Big Bang $\eta = 0$: under the Milne $\to$ Minkowski embedding, the Big-Bang hypersurface $\eta = 0$ maps onto the future lightcone of the origin, and $\overline{D_{\mathcal{O}}} \cap \{\eta = 0\} = \emptyset$ implies $\overline{M_{\mathcal{O}}} \cap \partial I^+(0) = \emptyset$. Hence Theorem 2 applies to any doubly-bounded comoving diamond of an open FRW with $\eta_i > 0$, with no additional hypothesis on the algebraic side beyond the $k = 0$ Theorem 3.5/3.6 of [12]. In this sense the $k = -1$ extension is folklore: it follows by direct composition with the classical isometry of [15, §5.3].

5 Theorem (closed FRW, $k = +1$): conditional type II_∞ + BMS partial resolution

For $k = +1$ the conformal target is the Einstein static universe $\tilde{g}_{+1} = g_{\mathrm{ESU}}$, which is *not* conformally embeddable into Minkowski as a single piece; the conformal compactification of Minkowski covers only *half* of ESU, and the spherical-diamond modular structure on full ESU does not reduce to Hislop–Longo on Minkowski by direct composition.

5.1 The conformal unitary, $k = +1$

Proposition 4 (Conformal unitary, $k = +1$)**.** *Under the hypotheses of Section 2.1 for $k = +1$ (smooth strictly positive $a(\eta)$ on a closed conformal-time interval $[\eta_i, \eta_f] \subset (0, \eta_*)$, bounded away from Big Bang $\eta = 0$ and Big Crunch $\eta = \eta_*$), and for $D_{\mathcal{O}} \subset (\mathrm{FRW}, k = +1)$ a doubly-bounded comoving causal diamond, there exists a unitary $U_{+1} : \mathcal{H}_{\mathrm{FRW}} \to \mathcal{H}_{\mathrm{ESU}}$ with*

(i) $U_{+1}\,\Omega_{\mathrm{FRW}} = \Omega_{\mathrm{ESU}}$*;*

(ii) $U_{+1}\,\varphi_{\mathrm{FRW}}(f)\,U_{+1}^{-1} = \tilde{\varphi}_{\mathrm{ESU}}(a^3 \cdot f)$ *for all* $f \in C_c^\infty(D_{\mathcal{O}})$*;*

(iii) $U_{+1}\,\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}}\,U_{+1}^{-1} = \mathcal{A}(D_{\mathcal{O}}^{\mathrm{ESU}})_{\mathrm{ESU}}$.

Here Ω_{ESU} denotes the conformally invariant ground state of the conformally coupled massless scalar on ESU.

Proof sketch. Identical to the proof of Proposition 1, with $\tilde{g}_{-1} = g_{\mathrm{Milne}}$ replaced by $\tilde{g}_{+1} = g_{\mathrm{ESU}}$ and using (5) for $k = +1$ (with $R[\tilde{g}_{+1}] = +6$). Existence of the conformally invariant ground state Ω_{ESU} of the conformally coupled scalar on ESU is classical (Dimock [5], Lüders–Roberts [10], and [15, §5.4]). □

5.2 Buchholz–Mund–Summers 2000: partial resolution

The closed-FRW analogue of Hislop–Longo on the spherical-diamond algebra $\mathcal{A}(D_{\mathcal{O}}^{\mathrm{ESU}})_{\mathrm{ESU}}$ of the conformally coupled scalar in Ω_{ESU} is not in the literature in the form needed here. Buchholz–Mund–Summers [2] establish a partial resolution. We quote their setup and results verbatim.

BMS setup. $\mathrm{RW} = (\mathbb{R} \times S^3, S^2(\tau)(\mathrm{d}\tau^2 - \mathrm{d}\sigma^2))$ with $\tau \in (-\alpha\pi/2, \alpha\pi/2)$, $0 < \alpha \leq 1$, conformally embedded in de Sitter via the diffeomorphism $\Phi(\tau, \chi, \theta, \phi) := (f(\tau), \chi, \theta, \phi)$, $f(\tau) := \arcsin(\sin(\alpha\pi/2)\sin\tau)$ (BMS eqs. 15–16). The transplantation $\Xi : \mathcal{W} \to \widetilde{\mathcal{W}}$ is defined on *wedges*

only (not on points), via $\widetilde{\mathcal{A}}(\widetilde{W}) := \mathcal{A}(W)$ for $\widetilde{W} = \Xi(W)$ (BMS eq. 53), and $\widetilde{\omega}(A) := \omega(A)$ (BMS eq. 54).

Theorem 5 (BMS Theorem 3.1, p. 14, verbatim)**.** *With the above definitions, the net $\{\widetilde{\mathcal{A}}(\widetilde{W})\}$ and state $\widetilde{\omega}$ satisfy the CGMA (Condition of Geometric Modular Action) and the Modular Stability Condition. The corresponding net of von Neumann algebras $\{\widetilde{\mathcal{R}}(\widetilde{W})\}$ satisfies wedge duality and transforms covariantly under $SO(4,1)$ under the adjoint action of $\mathcal{J}$. For $\alpha < 1$, there exist $J \in \mathcal{J}$ which do not implement point transformations of RW.*

Corollary 6 (BMS Corollary 3.2, $\alpha = 1$)**.** *For $\alpha = 1$, $\mathcal{J}$ represents the conformal group $SO(4,1)$ of RW $\simeq$ dS.*

Corollary 7 (BMS Corollary 3.3, double-cone non-triviality)**.** *Let $\widetilde{O}_{x,\varepsilon} \subset$ RW be a double cone of conformal-time half-extent ε centred at x. If $\varepsilon > \frac{\pi}{2}(1-\alpha)$, then $\widetilde{\Omega}$ is cyclic for $\widetilde{\mathcal{R}}(\widetilde{O}_{x,\varepsilon})$. If $\varepsilon < \frac{\pi}{2}(1-\alpha)$, then $\widetilde{\mathcal{R}}(\widetilde{O}_{x,\varepsilon}) = \mathbb{C} \cdot \mathbf{1}$ (trivial).*

What BMS resolves and what it leaves open. BMS [2] establish, on wedges of the transplanted RW net: (a) CGMA, (b) cyclic-separating $\widetilde{\Omega}$, (c) modular stability + $SO(4,1)$ covariance (Theorem 5); and (d) cyclic $\widetilde{\Omega}$ on diamonds of extent $\varepsilon > \frac{\pi}{2}(1-\alpha)$ (Corollary 7). They do *not* establish (e) type III_1 (the string "III_1" does not appear), (f) full Connes spectrum $\Gamma(\sigma^{\text{diamond}}) = \mathbb{R}$, or (g) the conformally coupled scalar concretely on ESU — BMS work with *abstract* CGMA-nets, citing the conformally coupled scalar ([5, ref. 12], [10, ref. 22]) only as a candidate for future work. The closing paragraph of [2] (p. 16, last paragraph before References) reads, verbatim:

> "…it would also be interesting to determine if the net [12] [Dimock 1980] and adiabatic vacuum states [22] [Lüders–Roberts 1990] … satisfy the CGMA and the Modular Stability Condition for our choice of wedges $\widetilde{\mathcal{W}}$."

This is the explicit "open problem" flag from Buchholz–Mund–Summers themselves: even the CGMA / modular stability question for the conformally coupled scalar on RW is open in their concluding section, let alone the type-classification and Connes-spectrum questions.

5.3 Residual hypothesis and conditional theorem

Hypothesis 8 ($\mathrm{H1}_{+1}$, residual form)**.** The wedge net of the conformally coupled massless scalar on the Einstein static universe $\mathbb{R} \times S^3$, in its conformally invariant ground state Ω_{ESU}, has type III_1 factors on its spherical-diamond subalgebras with full Connes spectrum $\Gamma(\sigma^{\text{diamond}}) = \mathbb{R}$. Equivalently, the net of [5] combined with the Lüders–Roberts / adiabatic ground state of [10] satisfies

(i) factoriality and type III_1 on doubly-bounded spherical diamonds, and

(ii) $\Gamma(\sigma^{\text{diamond}}) = \mathbb{R}$.

The hypothesis is *strictly weaker* than "$\mathcal{A}(D_{\mathcal{O}}^{\text{ESU}})_{\text{ESU}}$ satisfies all of (a)–(f) above" because (a)–(d) are now *theorems* (BMS [2], Theorem 5 + Corollary 6 + Corollary 7) for the dS-transplanted net, with the $\alpha = 1$ limit yielding the conformal RW=dS case unconditionally. Only (e)–(f), the type-and-spectrum input, remain.

Theorem 9 (Closed FRW, $k = +1$, conditional on $\mathrm{H1}_{+1}$)**.** *Let φ be the conformally coupled massless scalar on closed FRW with smooth strictly positive $a(\eta)$ on the closed conformal-time interval $[\eta_i, \eta_f] \subset (0, \eta_*)$ (bounded away from Big Bang and Big Crunch), in the conformal vacuum Ω_{FRW}.*

Let $D_{\mathcal{O}} \subset (\mathrm{FRW}, k = +1)$ be a doubly-bounded comoving causal diamond with conformal-time extent $\Delta\eta := \eta_f - \eta_i$. Let σ^{FRW} denote the modular automorphism group of $(\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}}, \Omega_{\mathrm{FRW}})$.

Then there exists a unitary $U_{+1} : \mathcal{H}_{\mathrm{FRW}} \to \mathcal{H}_{\mathrm{ESU}}$ with $U_{+1}\Omega_{\mathrm{FRW}} = \Omega_{\mathrm{ESU}}$ (Proposition 4), and:

1. ***(Wedge level, unconditional via BMS [2].)*** *The transplanted wedge algebra $\widetilde{\mathcal{R}}(\widetilde{W}_{\mathcal{O}})$ associated to $D_{\mathcal{O}}$ via the BMS transplantation Ξ satisfies geometric modular action (CGMA), modular stability and $SO(4,1)$ covariance with cyclic-separating $\widetilde{\Omega}$. For $\alpha = 1$ (conformal RW $\simeq$ dS), $\mathcal{J}$ implements the full $SO(4,1)$ conformal group of ESU.*

2. ***(Diamond level, unconditional via BMS Cor. 3.3.)*** *Provided $\Delta\eta > \frac{\pi}{2}(1-\alpha)$, $\widetilde{\Omega}$ is cyclic for the diamond algebra $\widetilde{\mathcal{R}}(\widetilde{D}_{\mathcal{O}})$.*

3. ***(Crossed-product type, conditional on H1$_{+1}$.)*** *Conditional on Hypothesis 8, $\sigma_t^{\mathrm{FRW}} = \mathrm{Ad}\, U_{+1}^{-1} \circ \sigma_t^{\mathrm{ESU}} \circ \mathrm{Ad}\, U_{+1}$, $\Gamma(\sigma^{\mathrm{FRW}}) = \mathbb{R}$, and $\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}} \rtimes_{\sigma^{\mathrm{FRW}}} \mathbb{R}$ is a type II_∞ factor.*

Proof. Items (1)–(2) follow from Proposition 4 together with the verbatim statements of Theorem 5 and Corollary 7 applied to the transplanted wedge / diamond algebra. Item (3) follows by the same Tomita–Takesaki + Connes–Takesaki argument as Theorem 2, using Ω_{ESU} in place of Ω_{Milne}: Hypothesis 8 delivers the type III_1 classification and the full Connes spectrum on $\mathcal{A}(D_{\mathcal{O}}^{\mathrm{ESU}})_{\mathrm{ESU}}$ that are needed to apply the Connes–Takesaki crossed-product type theorem [3, 4], and the W*-isomorphism invariance of Γ [12, Prop. 3.4] transports both back to $\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}}$. □

Remark 10. Items (1)–(2) of Theorem 9 are unconditional in the sense that they require no hypothesis beyond the BMS setup; they upgrade the "everything is conjectural" status of the closed-FRW companion theorem to "geometric modular action + modular stability + $SO(4,1)$ covariance + cyclicity on long-enough diamonds are theorems; only the type-classification of the crossed product remains conditional". The residual conditional input H1$_{+1}$ is narrowed from "the entire algebraic-quantum-field infrastructure" (where it stood before BMS) to "factoriality + type III_1 + Connes spectrum on diamonds" (where it stands after BMS).

6 Open questions

Open Question 11 (Unconditional H1$_{+1}$)**.** Prove or refute Hypothesis 8 unconditionally. Three strategies suggest themselves: (i) Brunetti–Guido–Longo [1] modular structure for diamonds in conformal QFT, together with the conformal embedding $\mathbb{R}^{1,3} \hookrightarrow$ ESU which covers exactly half of ESU, and an $O(2,4)$-conjugation argument transferring the spherical-diamond modular generator into the Minkowski half (folklore but not in the literature with the precision needed); (ii) explicit S^3 spherical-harmonic mode decomposition adapting Fröb [6] from de Sitter diamonds to ESU; (iii) extension of the BMS transplantation [2] from abstract CGMA-nets to the concrete Dimock–Lüders–Roberts net [5, 10] of the conformally coupled scalar.

Open Question 12 (Big Bang / Big Crunch double-singularity for $k = +1$)**.** The closed FRW recollapses at $\eta = \eta_*$ ($= \pi$ for radiation, 2π for matter) in addition to the Big Bang at $\eta = 0$. Diamonds touching $\eta = \eta_*$ admit the same three obstructions as diamonds touching $\eta = 0$ (failure of the rescaling bijection, IR singularity of the conformal-vacuum two-point function, non-Hadamard boundary behaviour; see [12, Question 6.4]). The inductive-limit construction $\mathcal{A}_0(D_{\mathrm{BC}})_{\mathrm{FRW}} := \lim_{\eta_f \to \eta_*^-} \mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}}$ towards the Big Crunch may admit a PCT-symmetric counterpart of the Big-Bang conjecture in [12, Question 6.4].

Open Question 13 (Cosmological no-boundary / Hartle–Hawking states)**.** The conformal vacuum used throughout is the Bunch–Davies / adiabatic-conformal state. For closed FRW the no-boundary (Hartle–Hawking) cosmological state and the Linde / Vilenkin tunnelling candidates yield different ground states on $\mathcal{A}(D_{\mathcal{O}}^{\mathrm{ESU}})_{\mathrm{ESU}}$. The type classification of the resulting modular crossed product, even granting $\mathrm{H1}_{+1}$ for the conformal-vacuum state, is open — in particular whether the no-boundary state delivers type II_∞ or II_1 (by analogy with the Bunch–Davies / static-patch dichotomy of CLPW [3]).

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This note records the spatially-curved extension of the conformal-pullback framework of [12]: the curvature-corrected identity (5), the Milne-via-Minkowski composition of Section 4, and the BMS-conditional formulation of Section 5. The author thanks the authors of [1,2,4–8,10,13–15] on whose works this short observation builds.

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